

SMART HOSTEL MANAGEMENT SYSTEM (SHMS)

Flipped Course Project Report - Programming in Java

Platform: VITyarthi | Academic Evaluation

Course: Programming in Java

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Architecture: Pure Java SE (Single File)

Verification Status: 100% Tests Passed

1. Abstract

Managing university student accommodation requires reliable coordination between room allocations, bed occupancies, diverse room tiers (Standard and Deluxe AC), semester billing, and dues clearance. The **Smart Hostel Management System (SHMS)** is a console-based Java application developed to demonstrate core Object-Oriented Programming (OOP) principles. It implements abstraction and inheritance hierarchies, dynamic polymorphic tariff calculations, encapsulation, custom exception handling, and single-file data persistence.

2. Problem Statement & System Objectives

Manual hostel record keeping results in double bed allocations, billing inconsistencies across differing room tiers, and loss of revenue when students vacate without clearing dues. SHMS resolves these through automated capacity validation, polymorphic billing formulas, and dues clearance checks.

3. Object-Oriented Architecture

The system is organized around two primary inheritance hierarchies:

- Room Hierarchy:** Abstract class `Room` extended by `StandardRoom` and `DeluxeACRoom`. Each subclass overrides `calculateTotalFee(months)` to account for tier-specific utility and electrical tariffs.
- Person Hierarchy:** Abstract class `Person` extended by `Student` with attributes for registration number, department, room allocation, and fee balance.

4. Core OOP Principles Applied

OOP Concept	Application in SHMS
Abstraction	Abstract base classes <code>Room</code> and <code>Person</code> specify contract methods while concealing implementation details.
Inheritance	<code>StandardRoom</code> and <code>DeluxeACRoom</code> inherit attributes from <code>Room</code> ; <code>Student</code> inherits from <code>Person</code> .

Polymorphism	Dynamic method dispatch computes room fees based on concrete instance type (Standard vs Deluxe AC).
Encapsulation	Private fields accessed exclusively via validated getter and setter methods.
Exception Handling	Custom exceptions (HostelException, RoomFullException, InvalidDataException) guard business operations.

5. Automated Test Suite Results

Test #	Test Case	Expected Result	Actual Result	Status
1	Room & Student Catalog Loading	Parse entities from data file	6 rooms, 5 students loaded	PASS
2	Polymorphic Fee Calculation	Accurate tariff computation	Std: Rs.48000, Deluxe: Rs.105000	PASS
3	Room Allocation & Bed Counter	Increment bed count on allocation	R103 bed count incremented	PASS
4	Capacity Limit & Over-allocation Check	Reject allocation when room full	RoomFullException caught	PASS
5	Fee Payment & Balance Update	Deduct payment from pending dues	Balance updated accurately	PASS
6	Vacate Room With Dues Check	Block vacate if balance remains	InvalidDataException caught	PASS

Overall Score: 6 / 6 Tests Passed (100.0%)

6. Declaration of Originality

I declare that this project and its accompanying documentation are my original work submitted for the Programming in Java course on the VITyarthi platform. All code logic, OOP structures, and tests were written independently in compliance with academic guidelines.